

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, OCTOBER 2024

Paper Code: ES-201**Subject: Computational Methods****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Marks	CO										
1(a)	State Rolle's and Lagrange's Mean Value Theorem. Also show that Rolle's theorem is a special case of Lagrange's Mean Value Theorem.	[2.5]	CO1										
1(b)	An approximate value to π is given by 3.1428571 and its true value is 3.1415926 . Find the absolute and relative error.	[2.5]	CO1										
1(c)	Evaluate $\int_0^1 [1 / (1 + x^2)] dx$ up to four decimal places, using Simpson's $\blacksquare$ rule taking $n = 4$.	[2.5]	CO2										
1(d)	Evaluate $\int_{-1}^1 [1 / (1 + x^2)] dx$ by one-point, two-point and three-point Gaussian Quadrature formula.	[2.5]	CO2										
2(a)	Find minimum value of $f(x_1, x_2) = 0.5 x_1^2 + 2.5 x_2^2$, using Steepest descent method for three iterations taking starting point as $X_1 = (5, 1)^T$.	[5]	CO1										
2(b)	Given that $f(5) = 150$, $f(7) = 392$, $f(11) = 1452$, $f(13) = 2366$ & $f(17) = 5202$. Evaluate $f(9)$, using Newton's divided difference interpolation formula.	[5]	CO2										
3(a)	Find the root of the equation $x - e^{-x} = 0$ correct to two decimal places using secant method by taking initial approximations as 0 and 1.	[5]	CO1										
3(b)	Given a set of points for the function $f(x)$, evaluate $f(0.5)$ using Newton's forward interpolation formula: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$f(x)$</td><td>5</td><td>6</td><td>3</td><td>14</td></tr></table>	x	0	1	2	3	$f(x)$	5	6	3	14	[5]	CO2
x	0	1	2	3									
$f(x)$	5	6	3	14									
4(a)	Find minimum value of $f(x) = x^3 - 3x - 5$, using Newton's method for three iterations taking initial approximation as $x_1 = 2.5$.	[5]	CO1										
4(b)	A curve passes through the points (0, 18), (1, 10), (3, -18) and (6, 90). Find the slope of the tangent to the curve at $x = 2$, using Lagrange's interpolation formula.	[5]	CO2										

**** End of Question Paper ****